

DevHub — Database Schema & Relationships

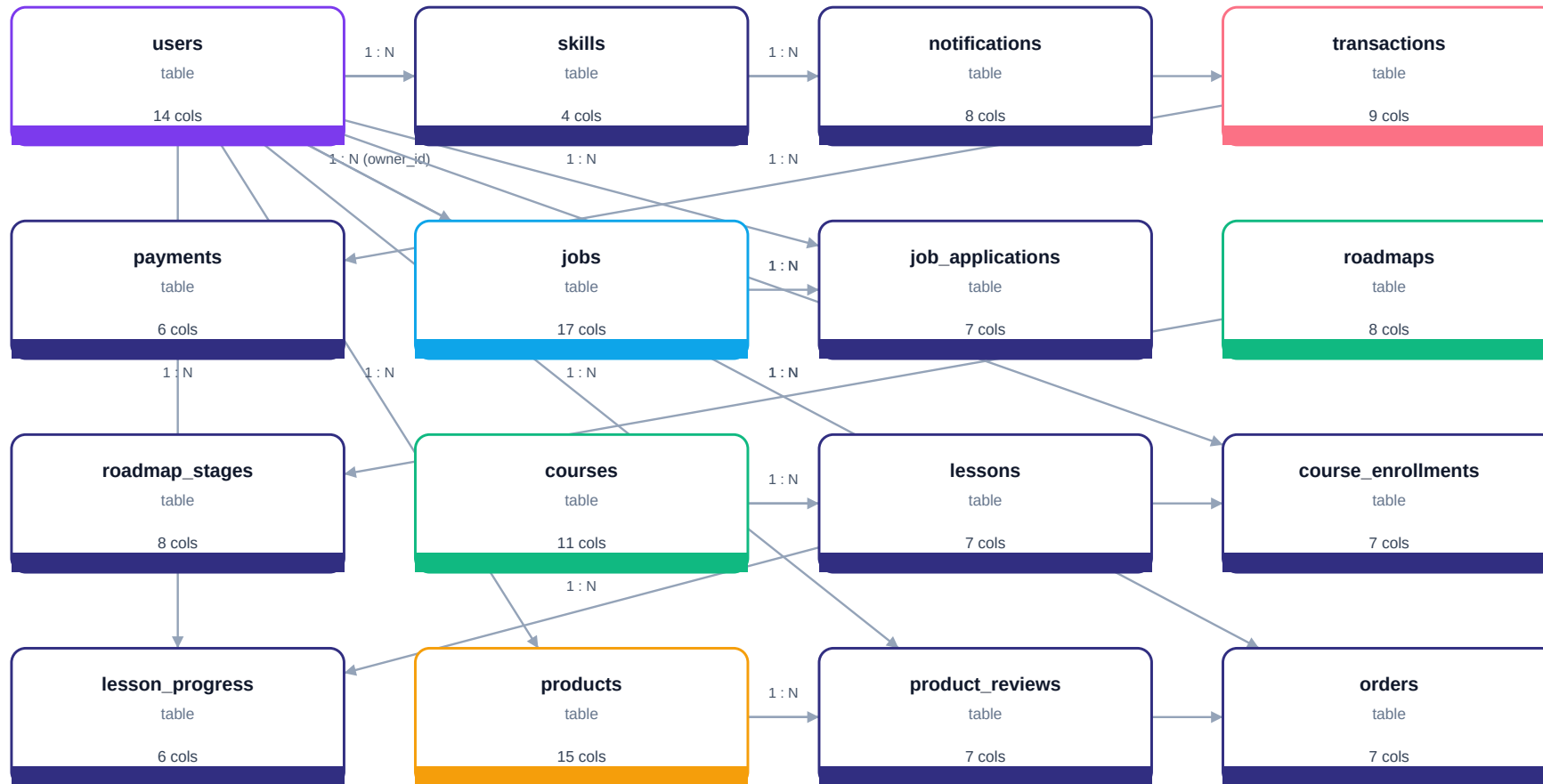
Reference document for the DevHub database structure: tables, columns, keys, and relationships between entities.

1. Overview

The backend uses the SQLAlchemy ORM. PostgreSQL is used by default and PostgreSQL in production. On the first startup all tables are created and sample (seed) data is inserted. Every table has an **id** column as its primary key.

Number of tables	16
Main entities	User, Job, Course, Product, Order, Transaction
Main areas	Auth & Users Jobs Learning Market Payments Notifications
Relationship logic	1:N via foreign keys, N:M via junction tables (course_enrollments and lesson_progress)

2. Relationships (ER diagram)



The lines between tables represent relationships; the label on each line states the relationship type. The junction tables (`course_enrollments`, `lesson_progress`) resolve many-to-many relationships and use UNIQUE constraints to prevent duplicates.

3. Table details

1. users — Users (developer, employer, admin)

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT, NOT NULL	Unique identifier
email	String(255)	UNIQUE, NOT NULL, index	Login email
full_name	String(255)	NOT NULL	Full name
password_hash	String(512)	NOT NULL	Password hash
role	String(32)	default 'developer'	User role
bio	Text	NULL	Biography
avatar_url	String(512)	NULL	Profile image URL
location	String(255)	NULL	Location
is_active	Boolean	NOT NULL, default True	Active status
is_verified	Boolean	NOT NULL, default False	Identity verified
is_employer	Boolean	NOT NULL, default False	Is an employer?
company	String(255)	NULL	Company name
created_at	DateTime	default utcnow	Creation date
updated_at	DateTime	onupdate utcnow	Last updated

2. skills — User skills

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
user_id	Integer	FK -> users.id, ON DELETE CASCADE, index	User

Column	Type	Constraints	Description
name	String(100)	NOT NULL	Skill name
level	String(32)	default 'beginner'	Skill level

3. jobs — Job postings and freelance projects

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT, index	Identifier
owner_id	Integer	FK -> users.id, ON DELETE CASCADE, index	Employer / listing owner
title	String(255)	NOT NULL	Title
company	String(255)	NOT NULL	Company
description	Text	NOT NULL	Description
location	String(255)	NULL	Location
type	String(64)	default 'full_time'	Type (full_time/part_time/...)
mode	String(64)	default 'remote'	Mode (remote/hybrid/on_site)
level	String(64)	default 'mid'	Seniority level
salary_range	String(128)	NULL	Salary range
skills	Text	default ''	Skills (comma separated)
is_active	Boolean	NOT NULL, default True	Active?
is_featured	Boolean	NOT NULL, default False	Premium / featured?
budget	Float	NULL	Budget
views	Integer	default 0	View count
created_at	DateTime	default utcnow	Created
updated_at	DateTime	onupdate utcnow	Updated

4. job_applications — Applications submitted for job postings

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
job_id	Integer	FK -> jobs.id, ON DELETE CASCADE, index	Job posting
candidate_id	Integer	FK -> users.id, ON DELETE CASCADE, index	Applicant
status	String(64)	default 'applied'	Application status
cover_letter	Text	NULL	Message / cover letter
created_at	DateTime	default utcnow	Created
updated_at	DateTime	onupdate utcnow	Updated

5. roadmaps — Learning roadmaps

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
title	String(255)	NOT NULL	Title
subtitle	String(512)	NULL	Subtitle
description	Text	NULL	Description
category	String(64)	default 'backend', index	Category
color	String(16)	NULL	Display color
is_published	Boolean	NOT NULL, default True	Published?
created_at	DateTime	default utcnow	Created

6. roadmap_stages — Stages of each learning roadmap

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
roadmap_id	Integer	FK -> roadmaps.id, ON DELETE CASCADE, index	Roadmap
order	Integer	default 0	Order
title	String(255)	NOT NULL	Stage title
description	Text	NULL	Description
resources	Text	NULL	Resources
project	Text	NULL	Hands-on project
checkpoint	Text	NULL	Progress checkpoint

7. courses — Training courses

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
title	String(255)	NOT NULL	Title
description	Text	NOT NULL	Description
category	String(64)	index	Category
level	String(64)	default 'beginner'	Level
instructor_name	String(255)	NULL	Instructor
cover_url	String(512)	NULL	Course image
duration_hours	Float	default 0	Duration
is_free	Boolean	NOT NULL, default True	Free?
price	Float	default 0	Price
created_at	DateTime	default utcnow	Created

8. lessons — Lessons of each course

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
course_id	Integer	FK -> courses.id, ON DELETE CASCADE, index	Course
order	Integer	default 0	Order
title	String(255)	NOT NULL	Lesson title
content	Text	NULL	Content
video_url	String(512)	NULL	Video
duration_minutes	Integer	default 0	Duration (minutes)

9. course_enrollments — User course enrollments (many-to-many)

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
course_id	Integer	FK -> courses.id, ON DELETE CASCADE, index	Course
user_id	Integer	FK -> users.id, ON DELETE CASCADE, index	User
progress	Float	default 0	Progress percentage
completed	Boolean	NOT NULL, default False	Completed?
enrolled_at	DateTime	default utcnow	Enrolled at
UNIQUE(course_id, user_id)	—	Constraint	Duplicate enrollments are not allowed

10. lesson_progress — User progress on lessons (many-to-many)

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
lesson_id	Integer	FK -> lessons.id, ON DELETE CASCADE, index	Lesson
user_id	Integer	FK -> users.id, ON DELETE CASCADE, index	User
completed	Boolean	NOT NULL, default False	Completed?
completed_at	DateTime	NULL	Completion time
UNIQUE(lesson_id, user_id)	—	Constraint	Duplicate progress rows are not allowed

11. products — Tool marketplace products

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
seller_id	Integer	FK -> users.id, ON DELETE CASCADE, index	Seller
title	String(255)	NOT NULL	Title
description	Text	NOT NULL	Description
category	String(64)	default 'template', index	Category
price	Float	NOT NULL, default 0	Price
currency	String(16)	default 'USD'	Currency
cover_url	String(512)	NULL	Image
demo_url	String(512)	NULL	Demo
tags	Text	default ''	Tags
rating	Float	default 0	Rating
sales	Integer	default 0	Sales count
is_published	Boolean	NOT NULL, default True	Published?

Column	Type	Constraints	Description
created_at	DateTime	default utcnow	Created
updated_at	DateTime	onupdate utcnow	Updated

12. product_reviews — Product reviews and ratings

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
product_id	Integer	FK -> products.id, ON DELETE CASCADE, index	Product
user_id	Integer	FK -> users.id, ON DELETE CASCADE, index	User
rating	Integer	NOT NULL, 1..5	Rating
comment	Text	NULL	Comment
created_at	DateTime	default utcnow	Created at
UNIQUE(product_id, user_id)	—	Constraint	One review per user per product

13. orders — Product purchase orders

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
product_id	Integer	FK -> products.id, ON DELETE CASCADE, index	Product
buyer_id	Integer	FK -> users.id, ON DELETE CASCADE, index	Buyer
amount	Float	NOT NULL	Amount
currency	String(16)	default 'USD'	Currency
status	String(32)	default 'pending'	Status
created_at	DateTime	default utcnow	Created at

14. transactions — User financial transactions

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
user_id	Integer	FK -> users.id, ON DELETE CASCADE, index	User
amount	Float	NOT NULL	Amount
currency	String(16)	default 'USD'	Currency
status	String(32)	default 'pending'	Status
provider	String(64)	default 'stripe'	Payment provider
reference	String(255)	NULL, index	Reference code
description	Text	NULL	Description
created_at	DateTime	default utcnow	Created at

15. payments — Recorded payments

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
transaction_id	Integer	FK -> transactions.id, ON DELETE CASCADE, NOT NULL	Transaction
method	String(64)	default 'card'	Payment method
paid_at	DateTime	NULL	Payment time
receipt_url	String(512)	NULL	Receipt
metadata	Text	NULL (column name: metadata)	Additional details

16. notifications — User notifications

Column	Type	Constraints	Description
id	Integer	PK, AUTOINCREMENT	Identifier
user_id	Integer	FK -> users.id, ON DELETE CASCADE, index	User
kind	String(32)	default 'info'	Notification kind
title	String(255)	NOT NULL	Title
message	Text	NULL	Message
link	String(512)	NULL	Link
is_read	Boolean	NOT NULL, default False	Read?
created_at	DateTime	default utcnow	Created at